

Dr. Kaisa Kajala



Plants protect their roots from stress and harm through specific cells known as barrier cell types. These cells can make compounds like lignin (wood) or suberin (cork) and put them on, almost like clothes which reinforce the cells to withstand harm. In my own research, I hope to understand how the different barrier cell types in roots of chickpea and maize are genetically regulated, and how this has evolved in different plant lineages. I am also part of the BarrierFates consortium.

ON PREVIOUS EXPERIENCES

Recently I wrote a chapter in an interdisciplinary book about plasticity. In my case it is about how adaptable root barriers are. This was challenging but very rewarding, and I am excited to see it come out soon. I had one of my research articles highlighted in the National Geographic Netherlands. Putting this together was really good fun.

There have been a few things that were challenging for me. I was part of a diversity campaign called “I’m a scientist”, where we had to do a one-minute video explaining our scientific excitement and interest. It was really difficult because I had to transmit enthusiasm while being uncomfortable in front of the camera. Every time I do something that I’ve never done before, I go through massive internal rebellion. With some science communication stuff I feel this way. It’s a combined I-really-liked-it-and-I-absolutely-hated-it. I’m super proud that I did it.

Can researchers have an impact on student’s future careers?

ON THE IMPACT

I think that the biggest impact that I have had with my career is the people who I’ve trained; the courses I’ve taught, the people who have been touched by my mentorship. People who I’ve seen go to do other things and have different roles in society. The biggest impact is the people.

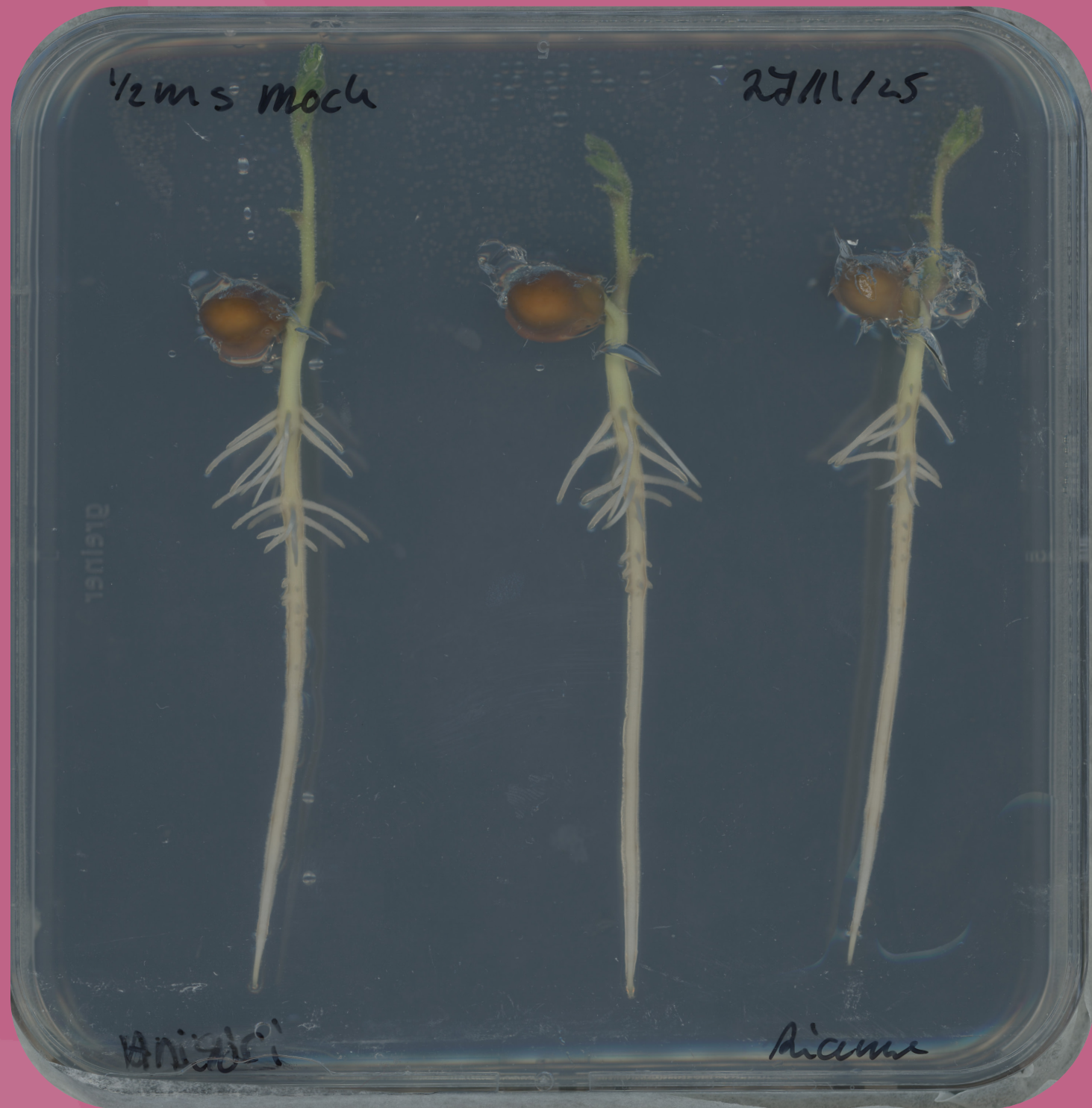
ON BARRIERFATES

Part of the BarrierFates outreach is this Fascination of Plants exposition. We wanted to bring the excitement for plant cell types, plant barriers, and plant resilience to the public. We will continue to have other events too, but this is the first of many to come.

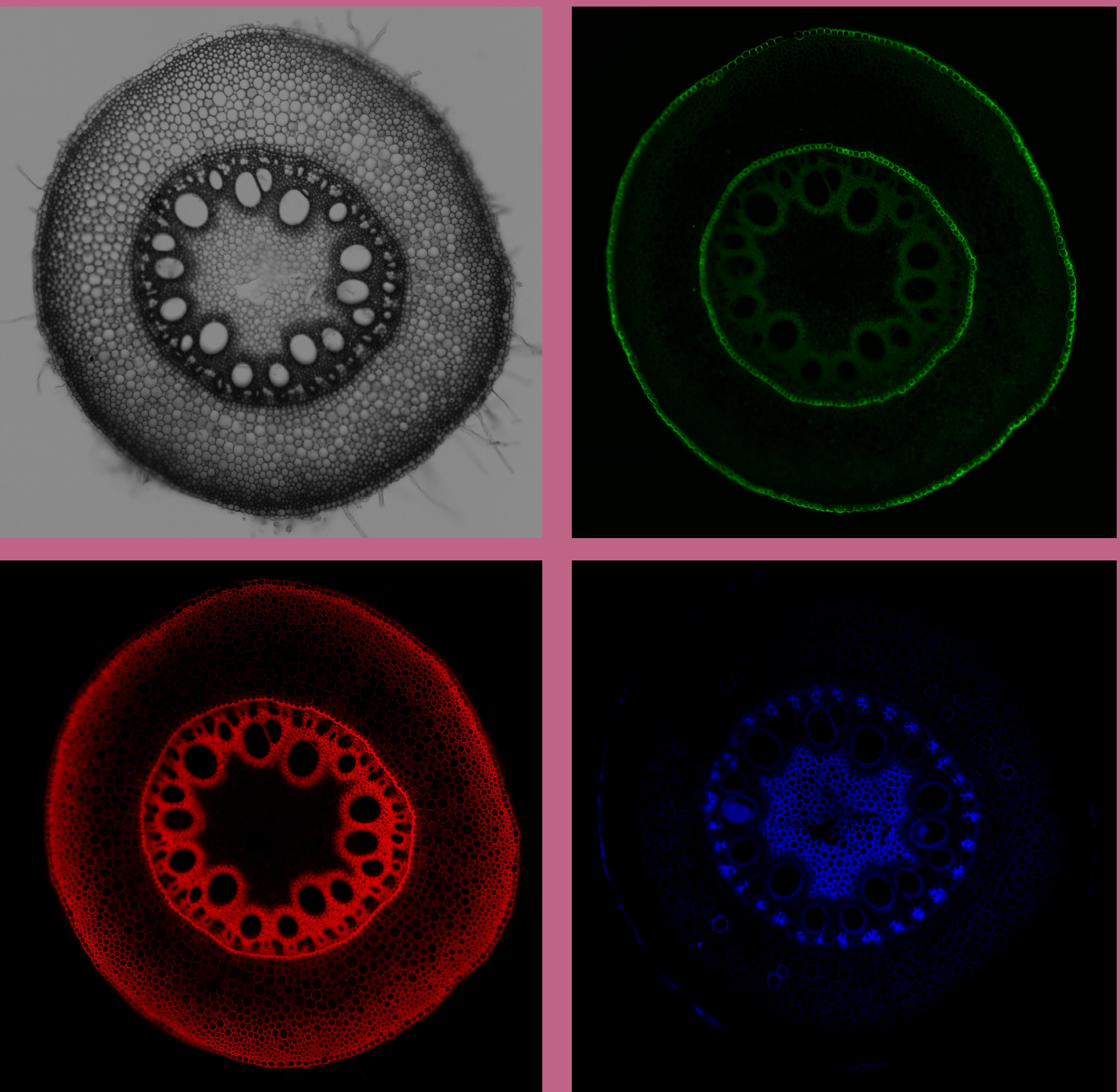
ON THE CHALLENGES

Recently, there’s been more encouragement to engage in science communication, but few real incentives. There are no consequences for opting out. At the same time, outreach is often expected, yet not always easily done. In the Netherlands, much of it happens in Dutch, which limits participation from non-native scientists.

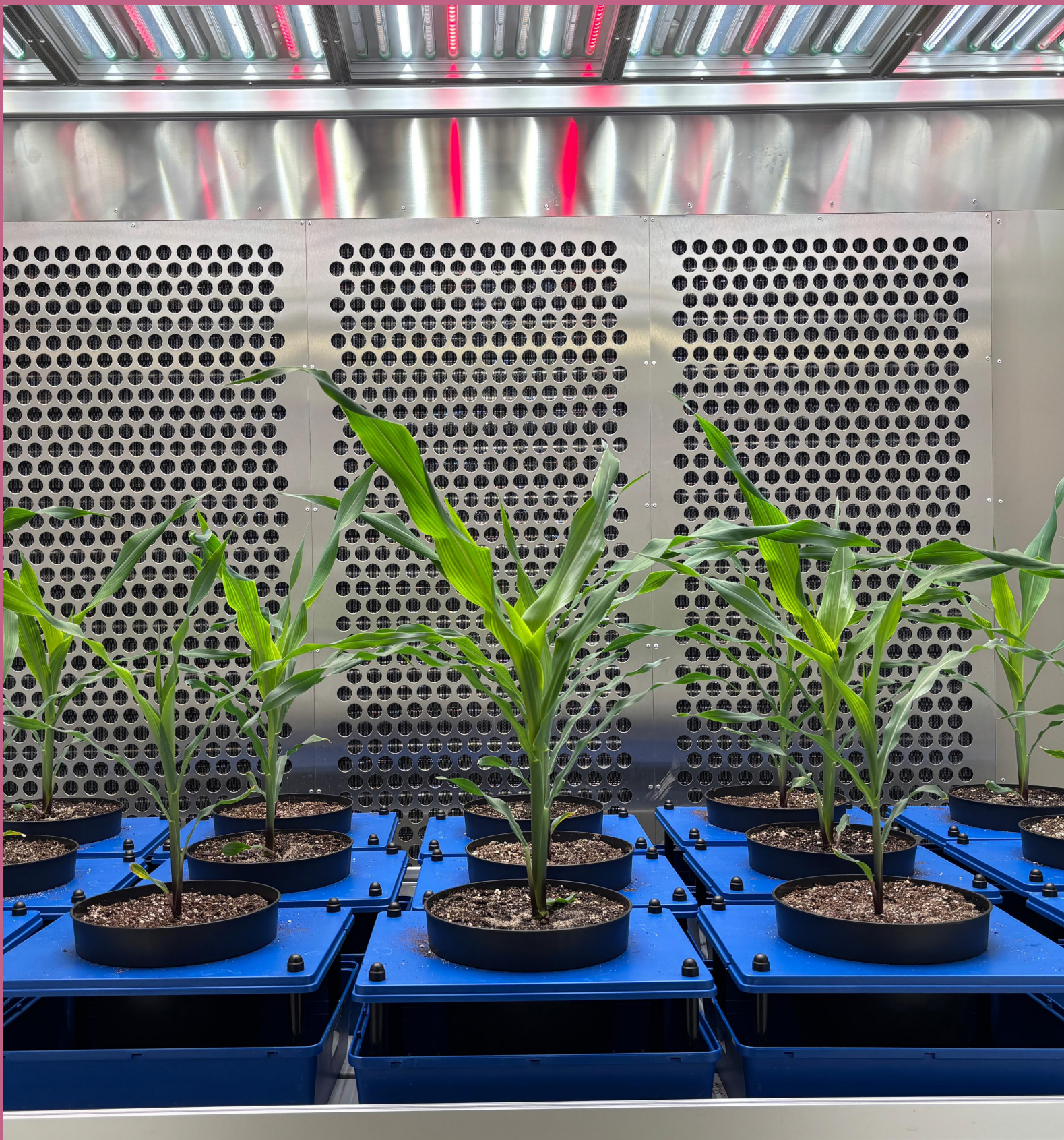
Conceptually, my research is easy to explain: plants with stronger roots can tolerate drought better. The complexity lies in identifying those plants, testing them across environments, and understanding their genetics for breeding. I haven’t been trained in science communication. I often assume people aren’t interested in the research, but only on the question and the findings at the end.



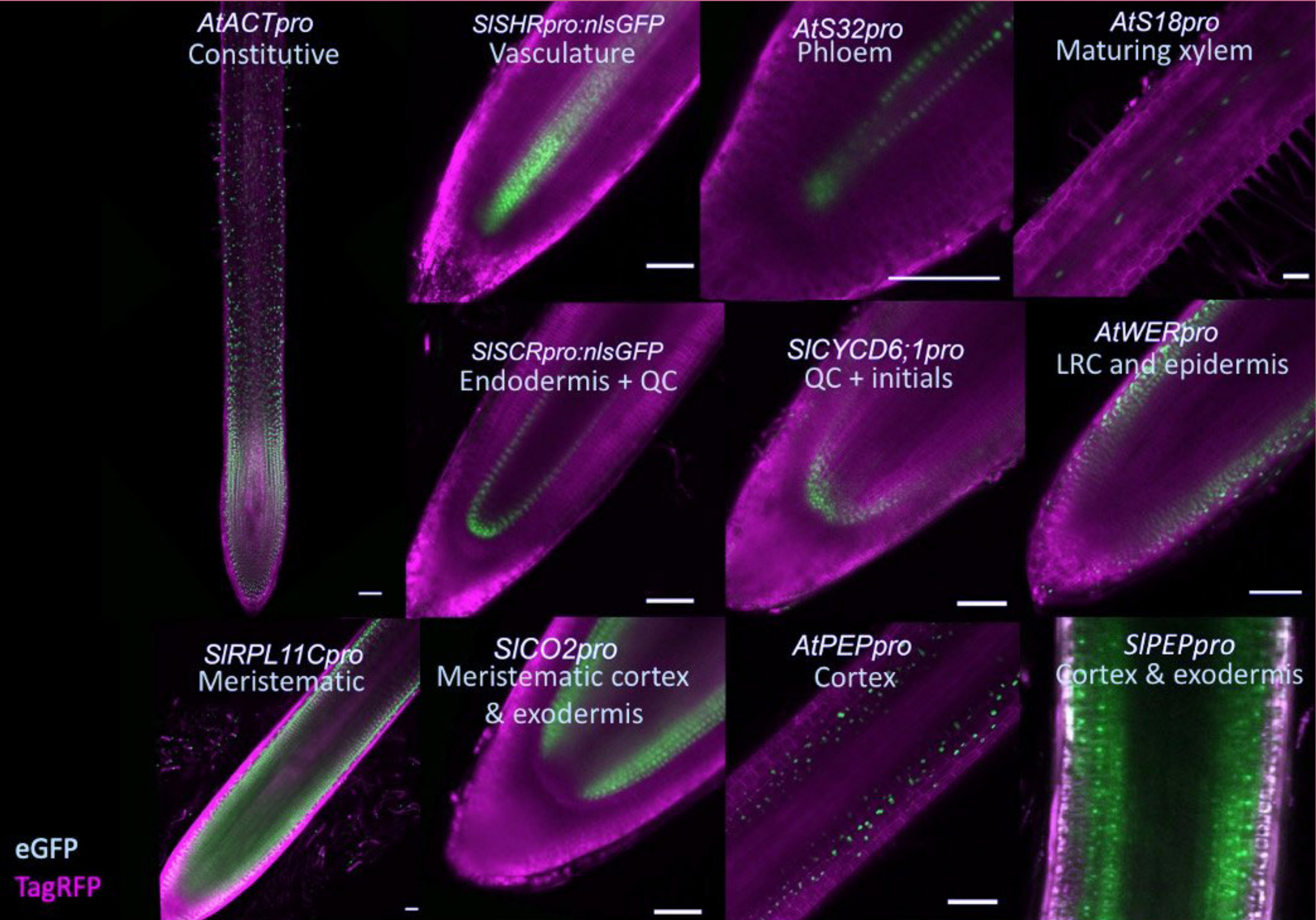
We grow chickpea plants on plates to study their root growth.



Maize roots can be sectioned, stained and viewed under the microscope to understand the composition of the cell walls.



Maize plants can be grown in a growth chamber to study their resilience to drought.



Genetic markers for tomato roots label specific cell types and allow studying molecular processes in a specific cell type. This allowed us to work out what genes are important for root barrier formation.

ON THE FUTURE

I think there are a few things we can do as scientists for science communication:

I’ve been thinking a lot about children’s programs now that my own kids are the age that we watch the Jeugdjournaal or Klokhuis every evening. It makes me think that we should have an episode about the science that we do!

Create roles or opportunities for outreach within job descriptions, especially for those interested, while encouraging and supporting all scientists to actively support it.

Read this poster in Dutch



What do you think?

